■■ Deep Learning Architectures & Their Purposes

Architecture	What it is	Main Uses	Examples
Feedforward Neural Networks (FNN / MLP)	Simplest NN, input → hidden → output.	Basic classification, regression on tabular data.	Basic MLP models
Convolutional Neural Networks (CNNs)	Uses filters to detect patterns in data.	Image recognition, computer vision, audio spectrograms.	AlexNet, ResNet, Efficient
Recurrent Neural Networks (RNNs)	Sequential models with memory of past inputs.	Time series, NLP (pre-transformer), speech.	LSTM, GRU
Transformers	Self-attention mechanism for sequences.	Language models, multimodal AI, vision.	GPT, BERT, ViT
Autoencoders	Compress & reconstruct data.	Dimensionality reduction, noise removal.	Denoising AE, Variational
Generative Adversarial Networks (GANs)	Generator vs Discriminator competition.	Image generation, data augmentation, deepfakes.	StyleGAN, BigGAN
Graph Neural Networks (GNNs)	Neural nets for graph-structured data.	Social networks, molecules, knowledge graphs.	GraphSAGE, GCN
Capsule Networks (CapsNets)	Preserve spatial relationships in vision	Advanced object recognition.	Experimental CapsNets